A Resource Theory of the Non-Hermitian Skin Effect

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Abstract

The non-Hermitian skin effect (NHSE), the accumulation of an extensive number of eigenmodes at a boundary, is usually diagnosed model by model through spectral winding. We formulate it as a quantum resource theory whose objects are the right and left eigen-responses of an arbitrary finite-dimensional generator together with a family of boundary collars, in any spatial dimension. The resource is *paired boundary localization*: right responses at a boundary and left responses at the opposite one. It is quantified by a skin score s that is the advantage of a boundary-only test in discriminating the two responses. We prove the following. (i) Every reciprocal generator ($G^T = UGU^\dagger$ with a boundary-compatible unitary U) with simple spectrum is exactly skin-free, and reciprocity with $U^T = -U$ forces degeneracy, which isolates the $\mathbb{Z}_2$ skin effect as the only reciprocal route to skin. (ii) Skin forces exponential spectral sensitivity: the eigenvalue condition number obeys $\log \kappa_m \geq s_m - \frac{1}{2}\log 2$, and the boundary-to-boundary residue of the resolvent is exponentially large and directional. (iii) The free operations apply one collar-non-expanding channel to every right and left response, whatever its mode, and process the mode label classically. They form a symmetric monoidal category, never create skin, and admit an infinite family of monotones, a faithful monotone, every data-processing divergence, and a complete family of monotones, computable by semidefinite programming, that characterizes exact and approximate convertibility. (iv) On generators, the free operations that keep the eigenmodes distinct are exactly the embeddings of a function of the generator as a decoupled block of a larger generator. They leave every eigenvalue condition number invariant. Non-unitary similarities (imaginary gauge fields) and one-way couplings are not free, and neither are reciprocal perturbations, which never create skin but can amplify it. (v) At the level of skin rates the theory is scalar: from a uniformly scaling object exactly the rates in $[0, r]$ are reachable. At fixed size, Hatano–Nelson chains with different couplings are mutually inconvertible, and the error of an approximate conversion is bounded below by an exponentially small collar deficit. (vi) For every tridiagonal generator with positive hopping products, whether clean, disordered or quasiperiodic, the skin score is bounded below by the accumulated imaginary gauge field minus an edge-amplitude penalty, which recovers the Hatano–Nelson rate. Beyond tridiagonal chains, lower bounds rest on an explicitly stated hypothesis imported from non-Bloch band theory.

1 Introduction

Under open boundary conditions, a non-Hermitian generator can localize an extensive fraction of its eigenvectors at a boundary. This is the non-Hermitian skin effect (NHSE) [1, 2, 3, 4, 5, 6, 7, 8], reviewed in [9, 10]. Its standard characterization is spectral: in one dimension, skin modes accompany a non-zero point-gap winding of the periodic-boundary spectrum [6, 7], and in higher dimensions a non-zero spectral area [8]. Its consequences are dynamical and operational: exponential sensitivity of the spectrum to perturbations [12, 14], large Petermann factors [15], and directional amplification [16, 17].

Quantum resource theories [21, 22] organize a physical property around three structures: free objects, free operations that cannot create the property, and monotones that quantify it and decide

which conversions are possible. This paper builds such a theory for the NHSE. Its design goals are as follows.

(U1) *Domain.* Every finite-dimensional generator with a boundary structure defines an object, in any dimension and for any lattice, disorder or internal structure.

(U2) *Null sector.* The free objects are exactly the objects with zero skin score, and they contain every Hermitian and every reciprocal generator with simple spectrum.

(U3) *Order.* The free operations never create skin, admit a *complete* family of monotones, and have a reading as physical operations on generators.

(U4) *Physics.* The resource controls measurable quantities (condition numbers, directional resolvent residues) and is calibrated on the standard models with explicit constants.

Sections 2–7 establish (U1)–(U4). Section 9 states precisely what is not established.

The structural input is that the objects are *pairs* of (classical–quantum) states, ω^R and ω^L, processed by a common channel. This makes the theory a restriction of the resource theory of asymmetric distinguishability [25] and of the comparison of statistical experiments [26, 27], so all of their monotones remain monotones here. The skin effect is the part of that distinguishability that can be read off at the boundary. It is isolated by one additional constraint on the free operations (collar non-expansion), and the specific physics enters through that constraint. The free operations are moreover *label-blind*: the same channel acts on every eigenmode, which is what an operation on the generator does; Section 5 identifies the free operations on generators.

2 Objects

Throughout, Σ is a finite set of boundary labels with a fixed-point-free involution ι that assigns to each boundary its opposite. In one dimension $\Sigma = \{+, -\}$ (right and left ends) and $\iota(\pm) = \mp$. In d dimensions $\Sigma = \{\pm x_1, \dots, \pm x_d\}$, and corners can be included. A width cutoff $W \in \mathbb{N}$ is fixed at each system size.

Definition 1 (Skin datum)**.** A skin datum X consists of:

- a finite-dimensional Hilbert space H;
- projectors $\Pi_{\sigma,w}$ on H for $\sigma \in \Sigma$ and $w \in [W]$, nested in w (the *collars*);
- a finite response set with weights $\pi_m > 0$;
- density operators ρ^R_m, ρ^L_m on H (the right and left responses).

X is *boundary-separated* if $\Pi_{\sigma,W}\Pi_{\iota\sigma,W} = 0$ for all σ, and then, by nesting, $\Pi_{\sigma,w}\Pi_{\iota\sigma,w} = 0$ for all w. The class of boundary-separated skin data is denoted $\mathfrak{D}$. The *type* of X is $(M, H, \{\Pi_{\sigma,w}\})$. The classical–quantum (cq) operators are $\omega^h_X = \sum_m \pi_m |m\rangle\langle m| \otimes \rho^h_m$ for $h \in \{R, L\}$.

Definition 2 (Generator datum)**.** Let G be a matrix on $H = \ell^2(\mathcal{C}) \otimes \mathbb{C}^d$, where $\mathcal{C}$ is a finite set of cells with open boundaries, and let $\Pi_{\sigma,w}$ project onto the w cell layers nearest boundary σ. For each simple eigenvalue λ_m, let R_m and L_m be unit vectors with $GR_m = \lambda_m R_m$ and $G^\dagger L_m = \bar{\lambda}_m L_m$, so that $L_m^\dagger R_m \neq 0$. The generator datum of G has responses $\rho^R_m = |R_m\rangle\langle R_m|$, $\rho^L_m = |L_m\rangle\langle L_m|$ and weights π (uniform unless stated otherwise; an energy window is also allowed). If G is Hermitian we take $L_m = R_m$, which also covers degenerate spectra. Liouvillian, Floquet and scattering generators enter the same way once a matrix with right and left eigen-responses is specified.

All boundary quantities below depend only on collar weights $\mathrm{Tr}[\Pi\rho]$, which are invariant under complex conjugation, so they do not depend on the convention $G^\dagger L = \bar{\lambda} L$. A POVM read-out $\{E_x\}$ of the boundary becomes projective under the isometry $\psi \mapsto \sum_x \sqrt{E_x}\psi \otimes |x\rangle$, and every fidelity and divergence below is invariant under a common isometry, so collars may be taken projective.

3 Skin score, skinless objects, and what skin forces

Definition 3 (Paired leakage and skin score)**.** For a response m, boundary σ and width w, the paired leakages are

$$\lambda_m^R(\sigma, w) = \mathrm{Tr}[(\mathbb{I} - \Pi_{\sigma,w})\rho_m^R], \qquad \lambda_m^L(\sigma, w) = \mathrm{Tr}[(\mathbb{I} - \Pi_{\iota\sigma,w})\rho_m^L].$$

The right response is tested at σ and the left response at the opposite boundary $\iota\sigma$. The skin score is

$$s_m(\sigma, w) = \tfrac{1}{2}\log^+ \frac{1}{\lambda_m^R(\sigma, w) + \lambda_m^L(\sigma, w)} \in [0, \infty], \qquad \log^+ x = \max(0, \log x),$$

and its average is $\mathcal{R}_\sigma(w) = \mathbb{E}_\pi\, s_m(\sigma, w)$.

Remark 1 (Operational meaning). Given a copy of ρ_m^R or ρ_m^L with equal priors, perform the boundary-only test $\{\Pi_{\sigma,w}, \Pi_{\iota\sigma,w}, \mathbb{I} - \Pi_{\sigma,w} - \Pi_{\iota\sigma,w}\}$. Answer "$R$" on the first outcome, "L" on the second, and abstain otherwise. The probability of a correct conclusive answer is $P_{\mathrm{succ}} = 1 - \frac{1}{2}(\lambda_m^R + \lambda_m^L)$, so

$$s_m = \tfrac{1}{2}\log^+ \frac{1}{2(1 - P_{\mathrm{succ}})}.$$

If the two responses coincide, $P_{\mathrm{succ}} \le \frac{1}{2}$. Thus $s_m > 0$ exactly when the boundary alone discriminates the right response from the left one better than it could discriminate two copies of the same state.

Definition 4 (Skinless objects)**.** $\mathcal{F} = \{X \in \mathfrak{O} :\ s_m(\sigma, w) = 0\ \forall m, \sigma, w\} = \{X \in \mathfrak{O} :\ \lambda_m^R + \lambda_m^L \ge 1\ \forall m, \sigma, w\}$.

Lemma 1 (Collar reciprocity implies skinless)**.** *If $X \in \mathfrak{O}$ and $\mathrm{Tr}[\Pi_{\sigma,w}\rho_m^R] = \mathrm{Tr}[\Pi_{\sigma,w}\rho_m^L]$ for all m, σ, w, then $X \in \mathcal{F}$.*

Proof. $\lambda_m^R + \lambda_m^L = 2 - \mathrm{Tr}[\Pi_{\sigma,w}\rho_m^L] - \mathrm{Tr}[\Pi_{\iota\sigma,w}\rho_m^L] \ge 1$, because $\Pi_{\sigma,w} + \Pi_{\iota\sigma,w}$ is a projector. □

Theorem 2 (Reciprocity)**.** *Let G be as in Definition 2, with transposition taken in a basis in which every $\Pi_{\sigma,w}$ is real. Suppose $G^T = UGU^\dagger$ for a unitary U that commutes with every $\Pi_{\sigma,w}$ (for example, a cell-local unitary).*

(a) For every simple eigenvalue λ_m, $\mathrm{Tr}[\Pi_{\sigma,w}\rho_m^R] = \mathrm{Tr}[\Pi_{\sigma,w}\rho_m^L]$ for all σ, w. Hence, if every response is a simple eigenvalue, the generator datum is skinless.

(b) If $U^T = -U$, then G has no simple eigenvalue.

Proof. From $G^T(UR_m) = UGR_m = \lambda_m UR_m$, the vector UR_m is an eigenvector of G^T. Moreover, $G^T v = \lambda v$ holds if and only if $G^\dagger \bar{v} = \bar{\lambda}\bar{v}$. (a) For a simple λ_m the left eigenspace is one-dimensional, so $L_m \propto \overline{UR_m}$. Since $\Pi = \Pi_{\sigma,w}$ is real and commutes with U, $\|\Pi\overline{UR_m}\| = \|\Pi UR_m\| = \|U\Pi R_m\| = \|\Pi R_m\|$. (b) If λ_m were simple, then $L_m \propto \overline{UR_m}$ as in (a), so $L_m^\dagger R_m \propto (UR_m)^T R_m = R_m^T U^T R_m$. This scalar equals its own transpose $R_m^T UR_m = -R_m^T U^T R_m$, so it vanishes. That contradicts $L_m^\dagger R_m \neq 0$. □

Corollary 3. *The following generator data are skinless: Hermitian generators, whatever their spectrum; complex-symmetric generators ($G^T = G$) with simple spectrum, such as reciprocal gain/loss chains; and all generators with simple spectrum that are reciprocal up to a cell-local gauge, $G^T = UGU^\dagger$ (then $U^T \neq -U$ by Theorem 2(b)).*

Remark 2 (The $\mathbb{Z}_2$ skin effect). Reciprocity with $U\bar{U} = -\mathbb{I}$ (equivalently $U^T = -U$) is the transpose-type time-reversal symmetry with $TT^* = -1$ [11], which hosts the $\mathbb{Z}_2$ skin effect [6]: Kramers-like partners accumulate at opposite ends. Theorem 2(b) shows that this degeneracy is *forced*, and part (a) that without it reciprocity kills skin, so degenerate eigenspaces are the only reciprocal route to skin. The datum then depends on a choice of biorthogonal basis in each eigenspace, and we make no claim about that case.

Theorem 4 (Skin forces spectral sensitivity)**.** *Let $X \in \mathfrak{O}$ and fix m, σ, w, abbreviating $\lambda^h = \lambda^h_m(\sigma, w)$ and $s = s_m(\sigma, w)$.*

(a) $F(\rho^R_m, \rho^L_m) \le \sqrt{\lambda^R} + \sqrt{\lambda^L} \le \sqrt{2}\, e^{-s}$, where $F(\rho, \tau) = \|\sqrt{\rho}\sqrt{\tau}\|_1$. Hence

$$\Phi_Q(X) := -\log \sum_m \pi_m F(\rho^R_m, \rho^L_m) \;\ge\; -\log \mathbb{E}_\pi e^{-s_m(\sigma,w)} - \tfrac{1}{2}\log 2.$$

(b) For generator data, let $P_m = R_m L_m^\dagger / (L_m^\dagger R_m)$ be the spectral projector of λ_m, and let $\kappa_m = \|P_m\| = |L_m^\dagger R_m|^{-1}$ be the eigenvalue condition number [13, 12]. Then

$$\log \kappa_m \;\ge\; s - \tfrac{1}{2}\log 2, \qquad \|\Pi_{\sigma,w} P_m \Pi_{\iota\sigma,w}\| = \kappa_m \sqrt{(1-\lambda^R)(1-\lambda^L)} \;\ge\; \sqrt{\tfrac{1}{2}(e^{2s}-1)},$$

and, if $s > 0$, $\|\Pi_{\iota\sigma,w} P_m \Pi_{\sigma,w}\| \le \frac{1}{2}\kappa_m e^{-2s}$.

Proof. (a) Because the collars are orthogonal, Remark 1 defines a valid three-outcome measurement. Its outcome laws are $p = (p_1, p_2, p_3)$ and $q = (q_1, q_2, q_3)$. Fidelity does not decrease under measurement [28], so $F \le \sum_i \sqrt{p_i q_i}$. Here $p_2 + p_3 = \lambda^R$ and $q_1 + q_3 = \lambda^L$. Cauchy–Schwarz gives $\sqrt{p_1 q_1} + \sqrt{p_3 q_3} \le \sqrt{(p_1+p_3)(q_1+q_3)} \le \sqrt{\lambda^L}$, and $\sqrt{p_2 q_2} \le \sqrt{\lambda^R}$. Next, $\sqrt{\lambda^R} + \sqrt{\lambda^L} \le \sqrt{2(\lambda^R + \lambda^L)}$, which equals $\sqrt{2}e^{-s}$ when $s > 0$; when $s = 0$ the bound $F \le 1 \le \sqrt{2}$ is trivial. Averaging over m gives the bound on Φ_Q. (b) For pure responses $F = |L_m^\dagger R_m| = \kappa_m^{-1}$, so the first inequality is (a). $\Pi_{\sigma,w} P_m \Pi_{\iota\sigma,w}$ has rank one with norm $\|\Pi_{\sigma,w} R_m\|\, \|\Pi_{\iota\sigma,w} L_m\|\, \kappa_m$, which gives the identity. Using $(1-\lambda^R)(1-\lambda^L) \ge 1 - e^{-2s}$ and $\kappa_m \ge e^s/\sqrt{2}$ gives the lower bound. For the reverse block, $\|\Pi_{\iota\sigma,w} R_m\|^2 \le \lambda^R$ and $\|\Pi_{\sigma,w} L_m\|^2 \le \lambda^L$, so it is at most $\kappa_m \sqrt{\lambda^R \lambda^L} \le \kappa_m(\lambda^R + \lambda^L)/2$. □

Theorem 4(b) is the physical content of the score. Near λ_m, $(z - G)^{-1} = P_m/(z - \lambda_m) + O(1)$, so $\Pi_{\sigma,w} P_m \Pi_{\iota\sigma,w}$ is the response at boundary σ to a drive at the opposite boundary. It exceeds $\sqrt{(e^{2s}-1)/2}$ and beats the reverse block by a factor of at least $2e^{2s}\sqrt{1 - e^{-2s}}$, which is the mechanism of directional amplification in driven-dissipative arrays [16, 17]. Since $|\delta\lambda_m| \le \kappa_m \|\delta G\| + O(\|\delta G\|^2)$ and κ_m^2 is the Petermann factor [15], a score s guarantees a sensitivity of order e^s [14]. For the Hatano–Nelson (HN) chain, $\log \kappa_j \simeq gL$ and $\max_{w \le L/2} s_j \simeq gL/2$, so the bound captures the exponential growth up to a factor of 2 in the rate.

Remark 3 (The converse fails). Non-normality does not imply skin. A Hermitian chain with one strongly non-reciprocal bulk bond (hoppings 3 and 0.2) has mean $\log \kappa_m \approx 0.68$ at every size but $\kappa_m \le 2.07$ uniformly in L, so Theorem 4(b) caps the score at $s_m < 1.1$ for all $w \le L/2$ and there is no accumulation (check V31-06, $L \le 160$). Skin is therefore a proper part of non-normality, and Theorem 4 is the one-way bridge between them.

Definition 5 (Directed skin accumulation)**.** A sequence (X_L) with widths w_L ($2w_L$ at most the linear size) shows *directed skin accumulation* at boundary σ and scale a (with $a(w) \to \infty$) if $\mathcal{R}_\sigma(w_L) \geq \eta\, a(w_L)$ for some $\eta > 0$ and all large L. The accumulation is *extensive* if moreover $\pi\{m : s_m(\sigma, w_L) \geq \eta\, a(w_L)\} \geq f > 0$. The rates are $r_\sigma^{\pm} = \limsup / \liminf_L \mathcal{R}_\sigma(w_L)/a(w_L)$. The sequence is *uniformly scaling with rate* r if $\max_m |s_m(\sigma, w_L)/a(w_L) - r| \to 0$.

4 Free operations and monotones

Definition 6 (Free processing and free morphism)**.** Let X have type (M, H, Π) and let $\tau = (M', H', \Pi')$ be a type. A *label-blind processing* of X into τ is the map

$$\mathcal{K} \otimes \Xi : \ |m\rangle\langle m| \otimes \rho \ \mapsto \ \textstyle\sum_{m'} \nu_{m'm} |m'\rangle\langle m'| \otimes \Xi(\rho)$$

on cq operators (the label is dephased first), where $\nu \in [0,1]^{M' \times M}$ is column-stochastic and $\Xi : B(H) \to B(H')$ is a CPTP map that is *collar non-expanding*:

$$\text{(C)} \quad \Xi^\dagger(\Pi'_{\sigma,w}) \leq \Pi_{\sigma,w} \qquad \text{for all } \sigma, w.$$

A *free processing* is a finite convex combination $\Theta = \sum_j p_j\, \mathcal{K}_j \otimes \Xi_j$ of label-blind processings into τ. A *free morphism* $X \to Y$ is a free processing into the type of Y such that

$$\text{(M1)} \quad \Theta(\omega_X^R) = \omega_Y^R, \qquad \Theta(\omega_X^L) = \omega_Y^L.$$

We write $X \to Y$ if a free morphism exists. Then $\pi^Y = \nu \pi^X$ with $\nu = \sum_j p_j \nu_j$.

Each ingredient has a physical reading. By (M1) the *same* processing acts on the right and left responses, blind to whether the forward problem G or the adjoint problem $G^\dagger$ was realized, as in the asymmetric-distinguishability setting [25]. By *label-blindness* one channel Ξ acts on every eigenmode, and the mode label is relabelled, merged or forgotten classically, with randomness independent of the mode. An operation on the generator acts on all its eigenmodes at once in this way, and Section 5 proves the converse. By (C), *collar non-expansion*, the probability of ending in a collar never exceeds that of having started there, so free operations never pump weight toward a boundary. This excludes non-reciprocal transport toward a boundary, the microscopic origin of skin. It also excludes reciprocal transport into a collar, which can move an existing right/left separation from width $w+1$ into width w and so raise the fixed-width score.

Proposition 5 (Examples of free morphisms)**.** *Each of the following is a label-blind free processing.*

(a) Boundary-compatible embeddings. $\nu = \mathbb{I}$ *and* $\Xi = V(\cdot)V^\dagger$ *for an isometry* $V : H \to H'$ *with* $V^\dagger \Pi'_{\sigma,w} V \leq \Pi_{\sigma,w}$*, such as a cell-local unitary, an enlarged internal space, inserted bulk cells, or cells attached beyond a boundary (Theorem 10).*

(b) Collar-compatible channels. $\nu = \mathbb{I}$ *and any* Ξ *satisfying (C), for example cell-local noise or dephasing.*

(c) Response processing. $\Xi = \mathrm{id}$ *and any column-stochastic* ν*.*

(d) Erasure. *If* ω *is a bulk state (*$\mathrm{Tr}[\Pi_{\sigma,W}\omega] = 0$ *for all* σ*), then* $\nu = (1, \ldots, 1)$ *and* $\Xi = \mathrm{Tr}(\cdot)\,\omega$ *give a free morphism* $X \to O_\omega$*, where* O_ω *has one response* $\rho^R = \rho^L = \omega$*. Moreover* $O_\omega \in \mathcal{F}$*.*

(e) Dilution. *For* $t \in [0,1]$*,* $\nu = \mathbb{I}$ *and* $\Xi_t = (1-t)\,\mathrm{id} + t\mathrm{Tr}(\cdot)\,\omega$ *give a free morphism* $X \to X_t$*, where* X_t *has responses* $(1-t)\rho_m^h + t\omega$*.*

Proof. Only (C) needs checking. In (a), $\Xi^\dagger(\Pi') = V^\dagger \Pi' V$; when the boundary is moved outward by k cells, $V^\dagger \Pi'_{\sigma,w} V = \Pi_{\sigma,w-k}$ (zero if $w \le k$). (b) is the hypothesis and (c) gives equality. In (d) and (e), the adjoint of $\mathrm{Tr}(\cdot)\,\omega$ is $A \mapsto \mathrm{Tr}[A\omega]\mathbb{I}$, which vanishes on collars. □

Lemma 6 (Leakage domination)**.** *Let $\Theta = \sum_j p_j \mathcal{K}_j \otimes \Xi_j : X \to Y$ be a free morphism, $\nu = \sum_j p_j \nu_j$, and $\beta_{mm'} = \nu_{m'm}\pi_m/\pi'_{m'}$, a probability vector in m. Then for all m', h, σ, w,*

$$\lambda^{h,Y}_{m'}(\sigma, w) \;\ge\; \sum_m \beta_{mm'} \lambda^{h,X}_m(\sigma, w).$$

Proof. Let Π be $\Pi_{\sigma,w}$ for $h = R$ and $\Pi_{\iota\sigma,w}$ for $h = L$, and Π' likewise. By (M1), $\pi'_{m'}\rho'^h_{m'} = \sum_{j,m} p_j \nu^j_{m'm} \pi_m \Xi_j(\rho^h_m)$. Hence, by (C),

$$\pi'_{m'}(1 - \lambda'^h_{m'}) = \textstyle\sum_{j,m} p_j \nu^j_{m'm} \pi_m \mathrm{Tr}[\Xi^\dagger_j(\Pi')\rho^h_m] \le \sum_m \nu_{m'm}\pi_m(1 - \lambda^h_m).$$

Divide by $\pi'_{m'} = \sum_m \nu_{m'm}\pi_m$. □

Conversely, a channel that violates (C) at (σ, w) lowers $\lambda^R(\sigma, w)$ of some one-response object. So (C) is the weakest condition on a label-blind channel under which every leakage monotone (Theorem 8(ii) below) is monotone.

Proposition 7 (Category)**.** *Skin data and free morphisms form a category. With the tensor product $X \otimes Y$ (Hilbert space $H_X \otimes H_Y$, responses (m, n) with weights $\pi_m\pi_n$, states $\rho^h_m \otimes \rho^h_n$, collars $\Pi^X_{\sigma,w} \otimes \Pi^Y_{\sigma,w}$) it is symmetric monoidal, and $\mathfrak{O}$ is closed under $\otimes$. The monoidal unit $\mathcal{I}$ (the space $\mathbb{C}$, one response, $\Pi_{\sigma,w} = 1$) is not boundary-separated. A morphism $\mathcal{I} \to X$ exists if and only if $\omega^R_X = \omega^L_X$.*

Proof. Composition: $(\mathcal{K}_2 \otimes \Xi_2)(\mathcal{K}_1 \otimes \Xi_1) = \mathcal{K}_2\mathcal{K}_1 \otimes \Xi_2\Xi_1$, and positivity of $\Xi^\dagger_1$ gives $(\Xi_2\Xi_1)^\dagger(\Pi'') \le \Xi^\dagger_1(\Pi') \le \Pi$. Composition is bilinear, so it preserves convex combinations, and (M1) composes. Tensor product: for $0 \le A \le A'$ and $0 \le B \le B'$, $A' \otimes B' - A \otimes B = (A' - A) \otimes B' + A \otimes (B' - B) \ge 0$, which gives (C) for $\Xi \otimes \Xi'$. Associators and the swap are label-blind and satisfy (C) with equality. Products of orthogonal projectors are orthogonal. For the unit, a free processing of $\mathcal{I}$ prepares a cq state, which (M1) forces to equal both ω^R_X and ω^L_X. Conversely, if $\omega^R_X = \omega^L_X$, preparing ρ^R_m with label m with probability π_m is a convex combination of label-blind processings, and (C) holds trivially because $\Pi^{\mathcal{I}}_{\sigma,w} = 1$. □

Theorem 8 (No skin creation, and monotones)**.** *Let $X \to Y$.*

(i) Golden rule. *If $X \in \mathcal{F}$ then $Y \in \mathcal{F}$.*

(ii) Leakage monotones. *Let $\psi : [0,1]^{2|\Sigma|W} \to (-\infty, \infty]$ be convex and non-increasing in each coordinate, and let $\Lambda_m = (\lambda^R_m(\sigma, w), \lambda^L_m(\sigma, w))_{\sigma,w}$. Then $\Psi_\psi(X) := \mathbb{E}_\pi \psi(\Lambda_m)$ satisfies $\Psi_\psi(Y) \le \Psi_\psi(X)$. This covers $\mathcal{R}_\sigma(w)$, the faithful monotone $\mathcal{S}(X) := \mathbb{E}_\pi \max_{\sigma,w} s_m(\sigma, w)$ (which vanishes if and only if $X \in \mathcal{F}$), and the rates of Definition 5 along levelwise free sequences.*

(iii) Divergence monotones. *For every function $\mathbf{D}$ on pairs of states that satisfies data processing, $\mathbf{D}(\omega^R_Y \| \omega^L_Y) \le \mathbf{D}(\omega^R_X \| \omega^L_X)$. Only (M1) is used. In particular Φ_Q is monotone and additive, $\Phi_Q(X \otimes Y) = \Phi_Q(X) + \Phi_Q(Y)$, and for generator data $\Phi_Q = -\log \mathbb{E}_\pi \kappa_m^{-1}$.*

Proof. (i) By Lemma 6, $\lambda'^R + \lambda'^L \geq \sum_m \beta_{mm'}(\lambda_m^R + \lambda_m^L) \geq 1$. (ii) Since ψ is non-increasing, Lemma 6 and Jensen's inequality give $\psi(\Lambda'_{m'}) \leq \psi(\sum_m \beta_{mm'}\Lambda_m) \leq \sum_m \beta_{mm'}\psi(\Lambda_m)$. Multiply by $\pi'_{m'}$, sum, and use $\sum_{m'} \nu_{m'm} = 1$. The score $s = \max\{0, -\frac{1}{2}\log(\lambda^R + \lambda^L)\}$ is a maximum of convex non-increasing functions, and so is $\max_{\sigma,w} s$. Faithfulness holds because all $\pi_m > 0$. (iii) Data processing under Θ together with (M1). For cq operators with a common classical part, $F(\omega^R, \omega^L) = \sum_m \pi_m F(\rho_m^R, \rho_m^L)$, and F is multiplicative under $\otimes$. □

Remark 4 (Copies versus size). Because Φ_Q is additive, exact conversion $X^{\otimes n} \to Y^{\otimes k}$ with $\Phi_Q(Y) > 0$ requires $k/n \leq \Phi_Q(X)/\Phi_Q(Y)$. The skin score is not additive (with intersection collars, $\lambda(X^{\otimes n}) = 1 - (1-\lambda)^n$), and the thermodynamic limit of the NHSE is system size, treated in Theorem 12, not copy number.

Theorem 9 (Completeness)**.**

(a) Leakage preorder. *For X, Y with finitely many responses the following are equivalent: (i) there is a column-stochastic ν with $\pi^Y = \nu\pi^X$ and $\Lambda^Y_{m'} \geq \sum_m \beta_{mm'}\Lambda^X_m$ coordinatewise (the conclusion of Lemma 6); (ii) $\Psi_\psi(Y) \leq \Psi_\psi(X)$ for all ψ as in Theorem 8(ii); (iii) the same holds for all $\psi(\Lambda) = \max_{k \leq M_Y}(\alpha_k - c_k \cdot \Lambda)$ with $c_k \geq 0$.*

(b) Full conversion. *For a type τ and Hermitian operators $U = (U_R, U_L)$ on $\mathbb{C}^{M'} \otimes H'$, let*

$$V_{\tau,U}(X) = \max\big\{\mathrm{Tr}[U_R\,\Theta(\omega_X^R)] + \mathrm{Tr}[U_L\,\Theta(\omega_X^L)] :\ \Theta \text{ a free processing of } X \text{ into } \tau\big\},$$

with $\max\emptyset = -\infty$. Each $V_{\tau,U}$ is a monotone. It is the value of a semidefinite program with one block per deterministic relabeling $f : [M] \to [M']$ (a linear program in the commuting sector). For Y of type τ, $X \to Y$ holds if and only if $V_{\tau,U}(X) \geq V_{\tau,U}(Y)$ for every U.

Proof. (a) (i)⇒(ii) is the proof of Theorem 8(ii), and (ii)⇒(iii) is trivial. For (iii)⇒(i), write (i) as a linear program in the coupling $J_{mm'} = \nu_{m'm}\pi_m \geq 0$ with constraints $\sum_{m'} J_{mm'} = \pi_m$, $\sum_m J_{mm'} = \pi'_{m'}$ and $\sum_m J_{mm'}\Lambda_m \leq \pi'_{m'}\Lambda'_{m'}$. If it is infeasible, Farkas' lemma [29] gives $a_m, b_{m'} \in \mathbb{R}$ and $c_{m'} \in \mathbb{R}^{2|\Sigma|W}_{\geq 0}$ with $a_m + b_{m'} + c_{m'} \cdot \Lambda_m \geq 0$ for all m, m' and $\sum_m \pi_m a_m + \sum_{m'} \pi'_{m'}(b_{m'} + c_{m'} \cdot \Lambda'_{m'}) < 0$. Set $\psi(\Lambda) = \max_{m'}(-b_{m'} - c_{m'} \cdot \Lambda)$. Then $a_m \geq \psi(\Lambda_m)$, and $\psi(\Lambda'_{m'}) \geq -b_{m'} - c_{m'} \cdot \Lambda'_{m'}$. Hence $\Psi_\psi(X) \leq \sum \pi_m a_m < -\sum \pi'_{m'}(b_{m'} + c_{m'} \cdot \Lambda'_{m'}) \leq \Psi_\psi(Y)$, which contradicts (iii).

(b) Every column-stochastic ν is a convex combination of deterministic relabelings f. Hence the free processings of X into τ are exactly the maps $\Theta = \sum_f \mathcal{K}_f \otimes \Phi_f$, where the Φ_f are CP maps with $\Phi_f^\dagger(\mathbb{I}) = t_f\mathbb{I}$, $\Phi_f^\dagger(\Pi'_{\sigma,w}) \leq t_f\Pi_{\sigma,w}$, $t_f \geq 0$ and $\sum_f t_f = 1$; given such Φ_f, $\Theta = \sum_{t_f>0} t_f\,\mathcal{K}_f \otimes (\Phi_f/t_f)$. These constraints are linear and semidefinite in the Choi operators of the Φ_f and define a compact convex set. Its image $\mathrm{Reach}_\tau(X) = \{(\Theta(\omega_X^R), \Theta(\omega_X^L))\}$ is therefore compact and convex, and $V_{\tau,U}$ is its support function, computed by the stated SDP. Composing a morphism $X \to X'$ with a free processing of X' into τ gives a free processing of X into τ (proof of Proposition 7), so $\mathrm{Reach}_\tau(X') \subseteq \mathrm{Reach}_\tau(X)$ and $V_{\tau,U}$ is monotone. The identity is a free processing of Y into its own type, so $V_{\tau,U}(Y) \geq \mathrm{Tr}[U_R\omega_Y^R] + \mathrm{Tr}[U_L\omega_Y^L]$. If $V_{\tau,U}(X) \geq V_{\tau,U}(Y)$ for all U, then (ω_Y^R, ω_Y^L) is not strictly separated from $\mathrm{Reach}_\tau(X)$, so it lies in that set, that is, $X \to Y$. The converse is monotonicity. □

Remark 5 (Approximate conversion). For any norm on pairs of operators, with dual norm $\|\cdot\|_*$, the minimal conversion error is

$$\min_\Theta \|\Theta(X) - Y\| = \max_{\|U\|_* \leq 1} \big[\mathrm{Tr}[U_R\omega_Y^R] + \mathrm{Tr}[U_L\omega_Y^L] - V_{\tau,U}(X)\big],$$

where Θ runs over free processings into the type τ of Y, by minimax over a compact convex set. It is again an SDP with all free-operation constraints imposed exactly, and each $V_{\tau,U}$ is the optimal payoff of a decision game played with boundary-constrained processing [23, 24]. The M'^M blocks are the price of label-blindness: mode-dependent channels would give a polynomial SDP, but they are not operations on generators (Remark 6(d)). For a single response and without (C), Theorem 9(b) reduces to the comparison of statistical experiments [26, 27].

5 Free operations on generators

Which operations on a generator induce free morphisms? Write $X(G)$ for the generator datum of G with all eigenvalues as responses (Definition 2), and $X_{\Lambda'}(G')$ for the datum of G' restricted to a set Λ' of simple eigenvalues. The weights are arbitrary. A generator is *irreducible* if no subspace other than 0 and H is invariant under both G and $G^\dagger$.

Theorem 10 (Free operations on generators)**.** *Let G on H have simple spectrum.*

(a) Let $V : H \to H'$ be an isometry with $V^\dagger \Pi'_{\sigma,w} V \le \Pi_{\sigma,w}$ for all σ, w, let h be injective on $\operatorname{spec} G$, and let $G' = Vh(G)V^\dagger + QKQ$ with $Q = \mathbb{I} - VV^\dagger$ and any K, such that the eigenvalues in $\Lambda' = h(\operatorname{spec} G)$ are simple for G'. Then $\mathcal{K}_f \otimes V(\cdot)V^\dagger$, where f relabels λ_m as $h(\lambda_m)$, is a free morphism $X(G) \to X_{\Lambda'}(G')$.

(b) Conversely, let $\Theta = \sum_j p_j \mathcal{K}_j \otimes \Xi_j$ be a free morphism $X(G) \to X_{\Lambda'}(G')$. Then every ν_j is a deterministic relabeling f_j. If f_j is injective, then:

(i) the span E'_j of the right eigenvectors of G' with labels in $f_j(\operatorname{spec} G)$ reduces G';

(ii) there are an isometry V_j of H onto E'_j and a function h_j, injective on $\operatorname{spec} G$, with $G'|_{E'_j} = V_j h_j(G) V_j^\dagger$;

(iii) $\Xi_j = V_j \mathcal{D}_j(\cdot) V_j^\dagger$, where $\mathcal{D}_j$ is a channel whose Kraus operators commute with G and $G^\dagger$. In particular $\Xi_j(\rho^h_m) = V_j \rho^h_m V_j^\dagger$, and $\mathcal{D}_j = \mathrm{id}$ if G is irreducible;

(iv) $\kappa_{f_j(m)}(G') = \kappa_m(G)$ and $s_{f_j(m)}(\sigma, w) \le s_m(\sigma, w)$ for all m, σ, w.

Proof. (a) Since $QV = 0$, $G'VR_m = Vh(G)R_m = h(\lambda_m)VR_m$, and likewise $G'^\dagger VL_m = \overline{h(\lambda_m)}VL_m$. As $h(\lambda_m)$ is simple for G', these are its unit eigenvectors, which gives (M1). (C) is the hypothesis on V.

(b) *Step 1.* Eigenvectors of distinct eigenvalues are linearly independent, so the target responses are pure and pairwise distinct. By (M1), $\pi'_{m'} \rho'^R_{m'} = \sum_{j,m} p_j \nu^j_{m'm} \pi_m \Xi_j(\rho^R_m)$. Pure states are extremal, so $\Xi_j(\rho^R_m) = \rho'^R_{m'}$ whenever $p_j \nu^j_{m'm} > 0$, and likewise for L. Hence each column of ν_j is supported on one label $f_j(m)$.

Step 2. Fix j with $f = f_j$ injective, and Kraus operators T_k of Ξ_j. Purity forces $T_k R_m = a_{km} R'_{f(m)}$ and $T_k L_m = b_{km} L'_{f(m)}$, with $\sum_k |a_{km}|^2 = \sum_k |b_{km}|^2 = 1$. Since the R_m span H, $\operatorname{ran} T_k \subseteq E'_R := \operatorname{span}\{R'_{f(m)}\}$, and likewise $\operatorname{ran} T_k \subseteq E'_L := \operatorname{span}\{L'_{f(m)}\}$. Every $R'_{f(m)}$ lies in some $\operatorname{ran} T_k$, so $E'_R \subseteq E'_L$, and by symmetry $E'_R = E'_L$. For the spectral projector P of G' onto the labels $f(\operatorname{spec} G)$, $\operatorname{ran} P = E'_R$ and $\operatorname{ran} P^\dagger = E'_L = (\ker P)^\perp$, so P is an orthogonal projector. It commutes with G' and, being self-adjoint, with $G'^\dagger$. So $E' := \operatorname{ran} P$ reduces G', which is (i).

Step 3. Let $G'' = \sum_m \lambda_m P'_{f(m)}$, where $P'_{m'}$ are the spectral projectors of G'. It is a function of $G'|_{E'}$, and evaluation on $\{R_m\}$ and $\{L_m\}$ gives $T_k G = G'' T_k$ and $T_k G^\dagger = G''^\dagger T_k$. Hence every $T_k^\dagger T_l$ commutes with G and $G^\dagger$.

Step 4. The $*$-algebra generated by G is $\bigoplus_i B(H_i) \otimes \mathbb{I}_{n_i}$ for an orthogonal decomposition $H = \bigoplus_i H_i \otimes \mathbb{C}^{n_i}$ [30], and simplicity of the spectrum forces $n_i = 1$. So the commutant is spanned by the projectors Z_i onto the H_i, each R_m and L_m lies in one block, and $T_k^\dagger T_l = \sum_i c_{kl}^{(i)} Z_i$. Fix i and k with $c_{kk}^{(i)} > 0$; then $V_i = T_k Z_i / (c_{kk}^{(i)})^{1/2}$ is an isometry on H_i. For R_n in block i, $T_l R_n$ and $V_i R_n$ are multiples of $R'_{f(n)}$, and the inner product with $T_k R_n$ fixes their ratio at $c_{kl}^{(i)}/(c_{kk}^{(i)})^{1/2}$, independent of n, so $T_l Z_i = \gamma_{li} V_i$. Block-diagonality of $T_k^\dagger T_l$ makes the ranges of the V_i orthogonal, so $V = \sum_i V_i Z_i$ is an isometry onto E' (by dimension), and $\Xi_j = V\mathcal{D}(\cdot)V^\dagger$, where $\mathcal{D}$ has Kraus operators $D_k = \sum_i \gamma_{ki} Z_i$. They commute with G and $G^\dagger$, and $\mathcal{D}$ fixes each ρ_m^h. As $VR_m \propto R'_{f(m)}$, $G'|_{E'} = Vh(G)V^\dagger$ with $h(\lambda_m) = \lambda'_{f(m)}$, which gives (ii) and (iii).

Step 5. An isometry preserves $|L_m^\dagger R_m|$. By (C), $\mathrm{Tr}[\Pi' V \rho_m^h V^\dagger] = \mathrm{Tr}[\Xi_j^\dagger(\Pi')\rho_m^h] \le \mathrm{Tr}[\Pi \rho_m^h]$ for every collar. This gives (iv). □

Theorem 10 identifies the free operations on generators. One that keeps the eigenmodes distinct can only embed a function $h(G)$ of the generator, through a boundary-compatible isometry (a cell-local change of basis, an enlarged internal space, inserted bulk cells, or cells attached beyond a boundary), as a decoupled block of a larger generator, and dephase between the symmetry sectors of G, which leaves the eigen-data unchanged. As h may be e^{-iGt} or a resolvent, stroboscopic and Green's-function data carry the same resource as G, and every condition number, hence every Petermann factor, is invariant. The other free operations (noise, erasure, dilution, merging of modes) act on recorded responses, not on generators.

Remark 6 (Operations that are not free).

(a) *Non-unitary similarities.* $G \mapsto SGS^{-1}$ sends R_m to SR_m but L_m to $S^{-\dagger}L_m$. The imaginary gauge transformation $S = \mathrm{diag}(e^{gn})$ takes the Hermitian chain ($\kappa_m = 1$) to $\mathrm{HN}_L(g)$, which has $\kappa_m > 1$. Theorem 10(b)(iv) excludes this for every free morphism that keeps the modes distinct.

(b) *One-way couplings.* For $G' = \left(\begin{smallmatrix} G & 0 \\ C & K \end{smallmatrix}\right)$ with disjoint spectra, $L'_m = (L_m, 0)$ and $R'_m \propto (R_m, x_m)$ with $x_m = (\lambda_m - K)^{-1} C R_m$, so $\kappa'_m = \kappa_m(1 + \|x_m\|^2)^{1/2}$. If some $x_m \neq 0$, the multiset of condition numbers strictly grows, which Theorem 10(b)(iv) excludes for free morphisms $X(G) \to X_{\mathrm{spec}\, G}(G')$ that keep the modes distinct. The spectral subspace of $\mathrm{spec}\, G$ is invariant but not reducing, so condition (i) of Theorem 10(b) fails as well.

(c) *Reciprocal perturbations.* Adding Δ with $\Delta^T = \Delta$ keeps a complex-symmetric generator complex-symmetric, so by Theorem 2 it never creates skin from such generators while the spectrum stays simple. It is nevertheless not free. For the chain adjacency matrix A and $0 < \delta < e^{-g}$, $\mathrm{HN}_L(g) - \delta A = c\,\mathrm{HN}_L(g_\delta)$ with $c = [(e^g - \delta)(e^{-g} - \delta)]^{1/2}$ and $g_\delta = \frac{1}{2}\log\frac{e^g - \delta}{e^{-g} - \delta} > g$. By Corollary 16, the faithful monotone $\mathcal{S}$ rises by at least $(g_\delta - g)W - C_L(g) - C_L(g_\delta)$, which is positive once $W/\log L$ is large. A reciprocal operation can therefore amplify skin that is already present, so never creating skin is much weaker than being free.

(d) *Mode-dependent channels.* If the channel could depend on the mode, $\rho_m \mapsto \rho_m \otimes |e_m\rangle\langle e_m|$ (into $H \otimes \mathbb{C}^M$, collars $\Pi \otimes \mathbb{I}$) would satisfy (M1) and (C) mode by mode, mapping $X(G)$ to the datum of $\sum_m \lambda_m P_m \otimes |e_m\rangle\langle e_m|$, which is not of the form in Theorem 10(a). No free processing does this when the R_m are pairwise non-orthogonal and the κ_m are not all equal, as for HN chains. Indeed, each ν_j is a relabeling f_j (Step 1), and f_j is constant, $f_j \equiv c_j$, since fidelity cannot decrease and distinct labels have orthogonal targets. Then $\kappa_{c_j}^{-1} = F(\rho'^R_{c_j}, \rho'^L_{c_j}) \ge \kappa_a^{-1}$ for all a, and every label is some c_j, so all κ_m would be equal.

6 Conversions

Proposition 11 (Dilution band)**.** *Let X_t be as in Proposition 5(e) with $t \le \frac{1}{2}$. Then $\lambda_m^h(X_t) = (1-t)\lambda_m^h(X) + t$, and*

$$\left| s_m(X_t) - \min\{s_m(X), \tfrac{1}{2}\log\tfrac{1}{2t}\} \right| \le \tfrac{1}{2}\log 2 \qquad \text{for all } m, \sigma, w.$$

Proof. Let $u = \lambda^R + \lambda^L$. Then $u' = (1-t)u + 2t$ satisfies $\frac{1}{2}\max(u, 2t) \le u' \le 2\max(u, 2t)$ for $t \le \frac{1}{2}$. Hence $-\frac{1}{2}\log u'$ lies within $\frac{1}{2}\log 2$ of $\min(-\frac{1}{2}\log u, \frac{1}{2}\log\frac{1}{2t})$. The map $x \mapsto x^+$ is 1-Lipschitz, and $[\min(x, c)]^+ = \min(x^+, c)$ for $c \ge 0$. □

Theorem 12 (Rate-level classification)**.** *Let (X_L) be uniformly scaling with rate r at boundary σ and scale a (Definition 5). (i) If $X_L \to Y_L$ for every L, then $r_{\sigma'}^{\pm}(Y) \le r_{\sigma'}^{\pm}(X)$ for every σ'. (ii) For every $c \in [0, r]$ there are free morphisms $X_L \to Y_L$ such that (Y_L) is uniformly scaling with rate c at σ. Hence the set of rates reachable from (X_L) is exactly $[0, r]$, and at the rate level the resource is totally ordered by the single number r.*

Proof. (i) is Theorem 8(ii) applied at each L. (ii) For $c = 0$ use erasure (Proposition 5(d)). For $c > 0$ dilute with $t_L = \frac{1}{2}e^{-2c\,a(w_L)}$. By Proposition 11, $|s_m(Y_L) - \min\{s_m(X_L), c\,a(w_L)\}| \le \frac{1}{2}\log 2$, so $s_m(Y_L)/a(w_L) \to \min(r, c) = c$ uniformly in m. □

At fixed size the order is much richer than at the rate level.

Theorem 13 (Hatano–Nelson chains are mutually inconvertible)**.** *Let $\mathrm{HN}_L(g)$ be the generator datum (one site per cell, uniform weights on all L modes) of the open chain with $G_{n+1,n} = e^g$ and $G_{n,n+1} = e^{-g}$. Let $L \ge 2$ and $1 \le W \le L/2$. If $g \ne g'$, then neither $\mathrm{HN}_L(g) \to \mathrm{HN}_L(g')$ nor $\mathrm{HN}_L(g') \to \mathrm{HN}_L(g)$. In particular $\mathrm{HN}_L(g)$ cannot be converted into the skinless Hermitian chain $\mathrm{HN}_L(0)$.*

Proof. Averaging Lemma 6 over π' shows that $X \to Y$ requires $c_Y^R(\sigma, w) \le c_X^R(\sigma, w)$ for all σ, w, where $c^R(\sigma, w) = \mathbb{E}_\pi \mathrm{Tr}[\Pi_{\sigma,w}\rho^R]$. By the similarity $G = SH_0S^{-1}$ with $S = \mathrm{diag}(e^{gn})$ (Section 7), mode j has cell law $p_j^{(g)}(n) \propto e^{2gn}\mu_j(n)$ with $\mu_j(n) = \sin^2(k_j n)$, $k_j = \pi j/(L+1)$, and $\mu_j(1) = \mu_j(L) = \sin^2 k_j > 0$. For $A = \{n \le w\}$ with $1 \le w \le L-1$, $\frac{d}{dg}p_j^{(g)}(A) = 2\,\mathrm{Cov}_g(n, \mathbf{1}_A) < 0$, because $\mathbf{1}_A$ is non-increasing in n and non-constant on the support. Hence $c^R(-, 1)$ is strictly decreasing in g and $c^R(+, 1)$ is strictly increasing. For $g > g'$ the conversion $\mathrm{HN}(g) \to \mathrm{HN}(g')$ violates the condition at $(-, 1)$. For $g < g'$ it violates it at $(+, 1)$. The reverse direction follows by exchanging g and g'. □

The obstruction is exact but can be exponentially weak:

Proposition 14 (Lower bound for approximate conversion)**.** *For every free processing Θ of X into the type of Y, and every h, σ, w,*

$$\|\Theta(\omega_X^h) - \omega_Y^h\|_1 \;\ge\; 2\,[c_Y^h(\sigma, w) - c_X^h(\sigma, w)]^+,$$

with $c^L(\sigma, w) = \mathbb{E}_\pi \mathrm{Tr}[\Pi_{\iota\sigma,w}\rho^L]$. For $\mathrm{HN}_L(g) \to \mathrm{HN}_L(g')$ with $g > g' > 0$, the right-hand side vanishes except in the orientation $\sigma = -$. There it is at most $2c_{g'}^h(-, W) \le \frac{(L+1)^2}{2(1-e^{-2g'})}e^{-2g'(L-W)}$ for $h = R, L$. For $g' = 0$ (the Hermitian target) it equals $2W/L$ up to exponentially small corrections, so the Hermitian chain is not even approximately reachable when $W \propto L$.

Proof. Lemma 6, averaged over π', gives $c^h_{\Theta(X)} \le c^h_X$, and $\|A\|_1 \ge 2|\mathrm{Tr}[(\mathbb{I}\otimes\Pi)A]|$ for traceless Hermitian A. For HN, $p_j^{(g)}$ is stochastically increasing in g (proof of Theorem 13), so for $g > g'$ only $\sigma = -$ contributes. There, $c^R_{g'}(-, W) = \mathbb{E}_j p_j^{(g')}(n \le W)$. The numerator is at most $\sum_{n\le W} e^{2g'(n-1)} \le e^{2g'(W-1)}/(1-e^{-2g'})$, and the normalization is at least $e^{2g'(L-1)} \sin^2 k_j \ge 4e^{2g'(L-1)}/(L+1)^2$. The left response is the mirror image. For $g' = 0$, $\sum_j \sin^2(k_j n) = \frac{L+1}{2}$ gives $c^R_0(-, W) = W/L$. □

For single HN modes in the commuting sector, the minimal total ℓ_1 error, computed by linear programming, equals the sum over h of these bounds to solver precision and decays exponentially in L (V31-09). Whether this also holds for the full quantum data is open; it would require an explicit coherent free processing.

7 Calibration: imaginary gauge fields and edge amplitudes

Theorem 15 (Gauge bound for tridiagonal generators)**.** *Let G be tridiagonal on $\ell^2(\{1,\dots,L\})$ with real diagonal V_n, and off-diagonals $t_n^+ = G_{n+1,n}$, $t_n^- = G_{n,n+1}$ satisfying $t_n^+ t_n^- > 0$. Put $\gamma_n = \frac{1}{2}\log(t_n^+/t_n^-)$ and $\Gamma_n = \sum_{k<n}\gamma_k$. Let H_0 be the real symmetric Jacobi matrix with diagonal V_n and off-diagonals $\mathrm{sgn}(t_n^+)\sqrt{t_n^+ t_n^-}$, and let ϕ_j be its unit eigenvectors. Then $G = SH_0S^{-1}$ with $S = \mathrm{diag}(e^{\Gamma_n})$, the spectrum is real and simple, $R_j \propto S\phi_j$, $L_j \propto S^{-1}\phi_j$, $\phi_j(1)\phi_j(L) \ne 0$, and for $1 \le w \le L/2$*

$$s_j(+, w) \;\ge\; \min\{\Gamma_w^+, \Gamma_w^-\} - \tfrac{1}{2}\log\frac{2}{\min\{\phi_j(1)^2, \phi_j(L)^2\}},$$

where $\Gamma_w^+ = \min_{n\le L-w}(\Gamma_L - \Gamma_n)$ and $\Gamma_w^- = \min_{n>w}(\Gamma_n - \Gamma_1)$. If $\gamma_n \ge \gamma > 0$ for all n, then $\Gamma_w^\pm \ge \gamma w$. The mirror statement holds for $\sigma = -$.

Proof. Conjugation by S turns the off-diagonals of H_0 into $t_n^\pm$. Since H_0 is Hermitian, $G^\dagger = S^{-1}H_0S$, so $G^\dagger S^{-1}\phi_j = \lambda_j S^{-1}\phi_j$. A Jacobi matrix with non-zero off-diagonals has simple spectrum, and its eigenvectors cannot vanish at an end, since the three-term recursion would then force $\phi_j = 0$. Now

$$\lambda_j^R(+, w) = \frac{\sum_{n\le L-w} e^{2\Gamma_n}\phi_j(n)^2}{\sum_n e^{2\Gamma_n}\phi_j(n)^2} \le \frac{e^{2(\Gamma_L - \Gamma_w^+)}\sum_n \phi_j(n)^2}{e^{2\Gamma_L}\phi_j(L)^2} = \frac{e^{-2\Gamma_w^+}}{\phi_j(L)^2},$$

because $\Gamma_n \le \Gamma_L - \Gamma_w^+$ for $n \le L - w$. Likewise $\lambda_j^L(+, w) \le e^{-2\Gamma_w^-}/\phi_j(1)^2$. Therefore $\lambda^R + \lambda^L \le 2e^{-2\min(\Gamma_w^+, \Gamma_w^-)}/\min(\phi_j(1)^2, \phi_j(L)^2)$. □

The skin score is thus the imaginary gauge flux Γ [1] accumulated toward the boundary, minus a penalty set by the edge amplitudes of the Hermitian partner H_0, for arbitrary real disorder and quasiperiodic potentials (V31-05). If H_0 is Anderson-localized, the edge amplitudes of states far from both ends are exponentially small, and the bound becomes a competition between bias and localization length. This is consistent with [1, 18], but we do not derive the critical point. The bound is empty for a single non-reciprocal bulk defect (Remark 3). Scale-free [20] and critical [19] skin modes have bounded s; classifying them would need a size-relative scale a.

Corollary 16 (Hatano–Nelson, two-sided)**.** *For $\mathrm{HN}_L(g)$ with $g > 0$, $L \ge 3$ and $1 \le w \le L/2$, uniformly in j,*

$$|s_j(+, w) - g\,w| \le C_L(g) := \tfrac{3}{2}\log(L+1) + g + \tfrac{1}{2}\log\frac{1}{1-e^{-2g}}, \qquad s_j(-, w) = 0.$$

If $w_L \le L/2$ and $w_L/\log L \to \infty$, the sequence is uniformly scaling with rate g at scale $a(w) = w$ and shows extensive directed skin accumulation with $f = 1$.

Proof. Here $\gamma_n = g$ and $\phi_j(n) = \sqrt{2/(L+1)}\sin(k_j n)$, so $\phi_j(1)^2 = \phi_j(L)^2 = \frac{2}{L+1}\sin^2 k_j \geq \frac{8}{(L+1)^3}$. Theorem 15 then gives $s_j \geq gw - \frac{3}{2}\log(L+1) + \log 2$.

For the upper bound, write $p(n) \propto e^{2g(n-1)}\sin^2(k_j n)$. The normalization is at most $e^{2g(L-1)}/(1-e^{-2g})$. With $n_0 = L - w - 1 \geq 1$,

$$p(n_0) + p(n_0+1) \geq e^{-2g(w+1)}(1-e^{-2g})\big[\sin^2(k_j n_0) + \sin^2(k_j(n_0+1))\big],$$

and the bracket equals $1 - \cos k_j \cos(k_j(2n_0+1)) \geq \frac{1}{2}\sin^2 k_j \geq \frac{2}{(L+1)^2}$. Hence $\lambda^R \geq 2e^{-2g(w+1)}(1-e^{-2g})/(L+1)^2$, which gives $s_j \leq g(w+1) + \log(L+1) + \frac{1}{2}\log\frac{1}{1-e^{-2g}} - \frac{1}{2}\log 2$.

For $\sigma = -$: $p^{(g)}$ is stochastically larger than $p^{(0)}$ (proof of Theorem 13), and the left law $q^{(g)}$ is the mirror image of $p^{(g)}$ under $n \mapsto L+1-n$, hence stochastically smaller than $p^{(0)}$, which is mirror-symmetric. Since $2w \leq L$, both $\{n > w\}$ and $\{n \leq L-w\}$ contain at least half of the mirror-symmetric weight. So $\lambda^R(-,w) = p^{(g)}(n > w) \geq p^{(0)}(n > w) \geq \frac{1}{2}$ and $\lambda^L(-,w) = q^{(g)}(n \leq L-w) \geq p^{(0)}(n \leq L-w) \geq \frac{1}{2}$. □

Hypothesis 1 (Exponential biorthogonal localization)**.** For a family of generators G_L there are $\gamma > 0$, $c \geq 0$, C and $f > 0$ such that, for a fraction at least f of the modes, $\lambda^R_m(\sigma,w) + \lambda^L_m(\sigma,w) \leq CL^c e^{-2\gamma w}$ for all $w \leq W_L$.

On the good modes $s_m \geq \gamma w - \frac{1}{2}\log(CL^c)$, so Hypothesis 1 gives extensive directed skin accumulation with $a(w) = w$, any $\eta < \gamma$ and any widths with $w_L/\log L \to \infty$. Theorem 15 establishes it for tridiagonal chains with bias $\gamma_n \geq \gamma > 0$ whenever a fraction f of modes has $\min\{\phi_j(1)^2, \phi_j(L)^2\} \geq L^{-c}$. For banded (block-)Toeplitz generators it is the content of non-Bloch band theory [2, 5, 4]: right eigenvectors decay with the generalized-Brillouin-zone radius, left eigenvectors decay oppositely, and a non-zero point-gap winding is equivalent to the NHSE [6, 7]. In higher dimensions the criterion is a non-zero spectral area [8]. We import these results without re-proving them. Once Hypothesis 1 holds, Sections 3–6 apply with no further assumption.

8 Numerical checks

The script `nhse_qrt_v31_checks.py` (numpy/scipy, about 15 s) tests the main statements independently; tolerances are stated in the script and its output is `nhse_qrt_v31_checks_output.txt`. Direct eigen-decompositions are used only at small sizes, where they agree with the exact similarity transformation to 10^{-13} (V31-05), or with condition numbers at most 10^6 (V31-02). At larger L we use the exact similarity transformation, because numerical eigensolvers fail at pseudospectral sensitivity $\sim e^{gL}$ [12]. The results are summarized in Table 1.

Not tested numerically: Theorem 9(b) in the non-commuting sector (this needs an SDP solver), the category axioms, and Hypothesis 1 beyond tridiagonal chains. Theorem 10(b) is tested through the dimensions of the Kraus and intertwiner spaces, not by optimizing over channels.

9 Discussion and scope

The results answer (U1)–(U4). The domain is every finite-dimensional generator with right and left eigen-responses and boundary collars (U1). The free objects are the objects with zero skin score. They contain all Hermitian and all reciprocal generators with simple spectrum, and the $\mathbb{Z}_2$ skin effect is traced to a forced degeneracy (U2). The free operations are blind to the right/left label and to the mode label, and never pump weight into a collar. They cannot create skin, they admit

Check	Statement	Result
V31-01	Lemma 1, Theorem 2	$s \le 3 \times 10^{-15}$; cell laws of R, L agree to 10^{-14}; $U^T = -U$: all eigenvalues degenerate
V31-02	Theorem 4	all bounds hold; residue identity to $3 \times 10^{-12}\kappa$; 2055 cases with $s > 1$
V31-03	Theorem 8(i),(ii)	6000 monotone tests, 0 violations; skinless→skinless 600/600
V31-04	Theorem 8(iii)	Φ_Q: 0 violations; additivity to 10^{-14}
V31-05	Theorem 15, Corollary 16	gauge bound slack ≥ 0.62 on 2.9×10^4 disordered and quasiperiodic cases; HN band up to $L = 400$
V31-06	Remark 3, controls	Hermitian, periodic HN: $s \le 10^{-14}$; bulk bond ($L \le 160$): $\kappa \le 2.07$, $s < 1$
V31-07	Theorem 9(a)	LP vs. Farkas certificate 300/300; separation 197/197
V31-08	Theorem 9(b), commuting sector	label-blind images reachable 25/25, mode-dependent images unreachable 25/25; separating $V_{\tau,U}$ 50/50; no violation on 25 reachable pairs
V31-09	Proposition 11, Theorem 13, Proposition 14	dilution band holds; inconvertibility 120/120; minimal LP error equals the lower bound ($4.4 \times 10^{-3} \to 1.2 \times 10^{-5}$, $L = 12 \to 24$)
V31-10	Theorem 10	embedding: eigen-data, (C), leakages, κ_m to 10^{-14}; dim of commutant 1 (HN), 8 (Hermitian), 2 (two blocks); admissible Kraus space = intertwiner space; non-reducing windows: none; $f(G)$: data unchanged
V31-11	Remark 6	$\kappa'_m/\kappa_m = (1 + \|x_m\|^2)^{1/2}$ to 10^{-14}; gauge field: $\kappa_m \ge 1.91$; $\mathrm{HN}(0.3) - 0.4A = c\,\mathrm{HN}(0.512)$, $\mathcal{S}$: $2.6 \to 4.8$, $14.6 \to 25.2$ ($L = 20, 100$); tagging excluded

Table 1: Numerical checks of the main statements (`nhse_qrt_v31_checks.py`).

explicit and complete monotones, and on generators they are exactly the boundary-compatible embeddings of functions of G (U3). The score bounds the condition number and the directional resolvent residue from below, and it reproduces the Hatano–Nelson rate with explicit constants, including disordered and quasiperiodic chains (U4).

The following is not established here. (1) *Beyond tridiagonal chains*, lower bounds rest on Hypothesis 1, imported from non-Bloch band theory for banded Toeplitz generators and unproved for general disordered or long-range ones. (2) *Degenerate spectra and exceptional points*: generator data use simple eigenvalues; at an exceptional point $\kappa_m = \infty$ and the construction must be modified. (3) *Maximality*: (C) is the weakest condition under which every leakage monotone is monotone (after Lemma 6), and operations that merely never create skin can amplify it (Remark 6(c)); whether reflections exchanging opposite boundaries should be free is left open. (4) *Merging relabelings*: Theorem 10(b) assumes that no two modes are merged; operations that merge modes act on recorded data only and are not characterized on generators. The SDP of Theorem 9(b) is exponential in the number of responses. (5) *Copies*: only the zero-error converse of Remark 4 is proved; asymptotic copy rates are not computed, and system size, not copy number, is the physical limit. (6) *Approximate conversion within the HN family* for full quantum data is open (Section 6). (7) *Interacting and many-body* generators fit Definition 1, but no calibration is given for them.

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